Technical Documentation

Engine Series 6R 183 TD13H
Base: Mercedes-Benz OM 447 hLA

Drive for Rail Vehicles

- Euro 2 -

Operating Instructions
M015243/01E

MTU
DETROIT DIESEL

Wichtig – Important – Importante

Bitte die Karte „Inbetriebnahmemeldung“ abtrennen und ausgefüllt an MTU Friedrichshafen GmbH zurücksenden.

Die Informationen der Inbetriebnahmemeldung sind Grundlage für den vertraglich vereinbarten Logistik-Support (Gewährleistung, Ersatzteile etc.).

Please complete and return the “Commissioning Note” card below to MTU Friedrichshafen GmbH.

The Commissioning Note information serves as a basis for the contractually agreed logistic support (warranty, spare parts, etc.).

Veuillez séparer la carte “Signalisation de mise en service“ et la renvoyer à la MTU Friedrichshafen GmbH.

Les informations contenues dans la signalisation de mise en service constituent la base pour l'assistance en exploitation contractuelle (garantie, rechanges, etc.).

Rogamos separen la tarjeta “Aviso de puesta en servicio“ y la devuelvan rellenada a MTU Friedrichshafen GmbH.

Las informaciones respecto al aviso de puesta en servicio constituyen la base para el soporte logístico contractual (garantía, piezas de repuesto, etc.).

Ritagliare “Avviso di messa in servizio“ e rispedirlo debitamente compilato alla MTU Friedrichshafen GmbH.

Le informazioni ivi registrate sono la base per il supporto logistico contrattuale (garanzia, ricambi, ecc.).

É gentileza cortar o cartão "Participação da colocação em serviço", preenché-lo e devolvé-lo a MTU Friedrichshafen.

Os dados referentes à colocação em serviço representam a base para o suporte logístico (garantia, peças sobressalentes, etc.) estabelecido contratualmente.

Postcard

MTU Friedrichshafen GmbH
Department MST
88040 Friedrichshafen
GERMANY

Bitte in Blockschrift ausfüllen!
Please use block capitals!
Prière de remplir en lettres capitales!
¡A rellenar en letras de imprenta!
Scrivere in stampatello!
Favor preencher com letras de forma!

Motornr.: Engine No.: N° du moteur: N° de motor: Motore N.: No. do motor:	Auftragsnr.: MTU works order No.: N° de commande: N° de pedido: N. commessa: No. do pedido:
Motortyp: Engine model: Type du moteur: Tipo de motor: Motore tipo: Tipo do motor:	Inbetriebnahmedatum: Date put into operation: Mise en service le: Fecha de puesta en servicio: Messa in servizio il: Data da colocação em serviço:
Eingebaut in: Installation site: Lieu de montage: Lugar de montaje: Installato: Incorporado em:	Schiffstyp / Schiffshersteller: Vessel/type/class / Shipyard: Type du bateau / Constructeur: Tipo de buque / Constructor: Tipo di barca / Costruttore Tipo de embarcação/estaleiro naval:

Endabnehmer/Anschrift: End user`s address: Adresse du client final: Dirección del cliente final: Indirizzo del cliente finale: Usuário final/endereço:
Bemerkung: Remarks: Remarques: Observaciones: Commento: Observações:

Inbetriebnahme-meldung

Commissioning Note

Notice de mise en service

Aviso de puesta en servicio

Avviso di messa in servizio

Participação da colocação em serviço

Notes for the user

This Manual consists of 11 main sections, marked by tab-index sheets.

These sections are listed on this page.

Each Section has its own detailed Table of Contents.

The contents is subdivided into Groups and Subgroups.

The Groups in the Main Sections B and G correspond to the Groups in the Spare Parts Catalogue and other Product Support publications.

Using the Table of Contents to trace an entry

- ↓ Table of Contents (general)
- ↓ Contents
- ↓ Group / Subgroup
- ↓ Heading

Using the Alphabetical Index to trace an entry

- ↓ Term in Alphabetic Index
- ↓ Group / Subgroup
- ↓ Heading

Publication Amendment Service

Please complete and return the Confirmation of Receipt card; you will then receive amendments to your manual.

Safety, Accident Prevention and Environmental Protection Instructions

These instructions must be read and followed by every person involved in operation, maintenance or transportation of the machinery plant.

General

In addition to the instructions in this publication, the applicable country-specific legislation and other compulsory regulations regarding accident prevention must be observed.

This MTU engine is a state-of-the-art product and conforms with all the applicable specifications and regulations. Nevertheless, persons and property may be at risk in the event of:

- Incorrect use
- Servicing, maintenance and repair carried out by untrained members of staff
- Modifications or conversions
- Noncompliance with the Safety Instructions

Correct use

The engine is intended exclusively for the application specified in the contract or upon delivery. Any other use is considered improper use. The manufacturer will accept no liability for any resultant damage. The responsibility is borne by the user alone.

Correct use also includes observation of and compliance with the maintenance specifications.

Personnel requirements

Work on the engine must only be carried out by properly qualified personnel.

The specified legal minimum age must be respected.

Responsibilities of the operating, maintenance and repair personnel must be specified.

Modifications or conversions

Modifications made by the customer to the engine may affect safety.

MTU will accept no liability or warranty claims for any damage caused by unauthorised modifications or conversions.

Organisational measures

This publication must be issued to all personnel involved in operation, maintenance, repair or transportation.

It must be kept at hand near the engine and accessible to all personnel involved in operation, maintenance, repair or transportation.

The personnel must be instructed on engine operation and repair by means of this publication, and in particular the safety instructions must be explained.

This is especially important for personnel who work on the engine only on an occasional basis. Such personnel must be given instructions repeatedly.

Spare parts

Only genuine MTU spare parts must be used to replace components or assemblies. In event of any damaged caused by the use of other spare parts, no liability or warranty claims vis-à-vis the engine manufacturer will be accepted.

Working clothes and protective equipment

Wear proper work clothing for all work.

Depending on the kind of work, use additional protective equipment, e.g. protective goggles, protective gloves, protective helmet, apron.

Work clothing must be tight-fitting so that it does not catch on rotating or projecting components.

Do not wear jewellery (e.g. rings, chains, etc.).

Transportation

Lift the engine only with the lifting eyes provided.

Use only the transportation and lifting equipment approved by MTU.

Take note of the engine centre of gravity.

The engine must only be transported in installation position.

In the case of special packaging with aluminium foil, suspend the engine on the lifting eyes of the bearing pedestal or transport with equipment suitable for heavy loads (forklift truck).

Prior to transporting the engine, it is imperative to install transportation locking devices for crankshaft and engine mounts.

Secure the engine against tilting during transportation. The engine must be especially secured against slipping or tilting when going up inclines and ramps.

Setting the engine down after transportation

Place the engine only on an even, firm surface.

Ensure appropriate consistency and load-bearing capacity of the ground or support surface.

Never place an engine on the oil pan unless expressly authorised by MTU on a case-by-case basis to do so.

Working with laser equipment

When working with laser equipment, always wear special laser-protection goggles.

Laser equipment can generate extremely intensive, concentrated radiation by the effect of stimulated emission in the range of visible light or in the infrared or ultraviolet spectral range. The photochemical, thermal and optomechanical effects of the laser can cause damage. The main danger is irreparable damage to the eyes.

Laser equipment must be fitted with the protective devices necessary for safe operation according to type and application.

For conducting ray procedures and measurement work, only the following laser devices must be used:
- Laser devices of classes 1, 2 or 3A,
- Laser devices of class 3B, which have maximum output only in the visible wavelength range (400 to 700 nm), a maximum output of 5 mW, and in which the beam axis and surface are designed to prevent any risk to the eyes.

Engine operation

When the engine is running, always wear ear protectors.

Ensure that the engine room is well ventilated.

Mop up any leaked or spilt fluids and lubricants immediately or soak up with a suitable bonding agent.

Exhaust gases from combustion engines are poisonous. Inhalation of poisonous exhaust gases is a health hazard. The exhaust pipework must be leak-free and must discharge the gases to atmosphere.

During engine operation, do not touch battery terminals, generator terminals or cables.

Inadequate protection of electrical components can lead to electric shocks and serious injuries.

When the engine is running, never release coolant, oil, fuel, compressed air or hydraulic lines.

Maintenance and repair

Compliance with maintenance and repair specifications is an important safety factor.

Unless expressly permitted, no maintenance or repair work must be carried out with the engine running. The engine must be secured against inadvertent starting and the battery disconnected. Attach sign "Do not operate" in operating area or to control equipment. Persons not involved must keep clear.

Never attempt to rectify faults or carry out repairs if you do not have the necessary experience or special tools required. Maintenance and repair work must only be carried out by authorised, qualified personnel.

Use only the proper, calibrated tools.

Do not work on engines or components which are only held by lifting equipment or crane. Always support these components in accordance with regulations on suitable frames or stands before beginning any maintenance or repair work.

Before barring the engine, make sure that nobody is standing in the danger zone. After working on the engine, check that all guards have been reinstalled and that all tools and loose components have been removed from the engine.

Fluids emerging under high pressure can penetrate clothing and skin and may cause serious injury. Before starting work, relieve pressure in systems and H.P. lines which are to be opened.

Never bend a fuel line and do not install bent lines. Keep fuel injection lines and connections clean. Always seal connections with caps or covers if a line is removed or opened.

During maintenance and repair work, take care not to damage the fuel lines. To tighten connections when installing lines, use the correct tightening torque and ensure that all retainers and dampers are installed correctly.

Ensure that all fuel injection lines and compressed oil lines have sufficient play to avoid contact with other components. Do not place fuel or oil lines near hot components, except when necessary for design reasons during installation.

Take care with hot fluids in lines, pipes and chambers ⇒ Risk of injury!

Note cooling period for components which are heated for installation or removal ⇒ Risk of injury!

Take special care when removing ventilation or plugs from engine. In order to avoid the discharge of highly pressurised liquids, hold a cloth over the screw or plug. It is even more dangerous if the engine has recently been shut down, as the liquids can still be hot.

Take special care when draining hot fluids. ⇒ Risk of injury!

When draining, collect fluids in a suitable container, mop up any spilt fluids or wipe or soak them up with a suitable bonding agent.

When changing engine oil or working on the fuel system, ensure that the engine room is adequately ventilated.

When working high on the engine, always use suitable ladders and work platforms. Make sure components are placed on stable surfaces.

In order to prevent back injuries when lifting heavy components, adults should not lift weights between max. 10 kg and 30 kg, depending on age and sex.

- Use lifting gear or seek assistance.
- Ensure that all chains, hooks, slings, etc. are tested and authorised, are sufficiently strong and that hooks are correctly positioned. Lifting eyes must not be unevenly loaded.

Welding work

Never carry out welding work on the engine or engine-mounted units.

Never use the engine as an earth connection. – (This prevents the welding current passing through the engine and causing scorching or burning at bearings, sliding surfaces and tooth flanks, which can lead to pitting or other material damage).

Never position the welding power supply cable adjacent to, or crossing MTU plant wiring harnesses. (The welding current could be induced in the cable harnesses which could possibly damage the electrical plant).

The welding unit earth connection must not be more than 60 cm from the weld point.

If components (e.g. exhaust manifold) are to be welded, they must first be removed from the engine.

It is not necessary to remove the connector and connections when carrying out welding operation on MTU electronics if the master switch for power supply is switched from "ON" to "OFF" and the wire is disconnected from the negative and positive poles on the battery.

Hydraulic installation and removal

Only the hydraulic installation and removal equipment specified in the work schedule and in the assembly instructions must be used.

The max. permissible push-on pressure specified for the equipment must not be exceeded.

The H.P. lines for hydraulic installation and removal are tested with 3800 bar.

Do not attempt to bend or apply force to lines.

Before starting work, pay attention to the following:
- Vent the hydraulic installation/removal tool, the pumps and the lines at the relevant points for the system to be used (e.g. open vent plugs, pump until bubble-free air emerges, close vent plugs).
- For hydraulic installation, screw on the tool with piston retracted.
- For hydraulic removal, screw on the tool with piston extended.

For a hydraulic installation/removal tool with central expansion pressure supply, screw spindle into shaft end until correct sealing is achieved.

During hydraulic installation and removal, ensure that nobody is standing in the immediate vicinity of the component to be installed/removed. As long as the system is under pressure, there is the risk that the component to be installed/removed may be suddenly released from the pressure connection.

Before use, the tools must be checked at regular intervals (crack test).

Working on electrical assemblies

Switch off all live appliances before carrying out any work on electrical assemblies.

Gases released from the battery are explosive. Avoid sparks and naked flames. Do not allow battery acids to come into contact with skin or clothing. Wear protective goggles. Do not place tools on the battery. Before connecting the cable to the battery, check battery polarity. Battery pole reversal may lead to injury through the sudden discharge of acid or bursting of the battery body.

Do not damage wiring during removal work and when reinstalling wiring and ensure that during operation it is not damaged by contact with sharp objects, by rubbing against another component or by a hot surface.

Do not secure wiring to fluid-carrying lines.

On completion of the maintenance and repair work, any cables which have become loose must be correctly connected and secured.

Always tighten connectors with a connector pliers.

If wires are installed beside mechanical components and there is a risk of chafing, use cable clamps to properly support the wires.

For this purpose, no cable straps must be used as, during maintenance and / or repair work, the straps can be removed but not installed a second time.

Operation of electrical equipment

When operating electrical equipment, certain components of this equipment are live.

Noncompliance with the warning instructions given for this equipment may result in serious injury or damage to property.

Fire prevention

Rectify any fuel or oil leaks immediately; even splashes of oil or fuel on hot components can cause fires – therefore always keep the engine in a clean condition. Do not leave cloths soaked with fluids and lubricants lying around on the engine. Do not store combustible fluids near the engine.

Do not weld pipes and components carrying oil or fuel. Before welding, clean with a nonflammable fluid.

When starting the engine with a foreign power source, connect the ground lead last and remove it first. To avoid sparks in the vicinity of the battery, connect the ground lead from the foreign power source to the ground lead of the engine or to the ground terminal of the starter.

Always keep suitable fire-fighting equipment (fire extinguishers) at hand and familiarise yourself with their use.

Noise

Noise can lead to an increased risk of accident if acoustic signals, warning shouts or noises indicating danger are drowned.

At all workplaces with a sound pressure level over 85 dB(A), always wear ear protectors (protective wadding, plugs or capsules).

Environmental protection

Dispose of used fluids and lubricants and filters in accordance with local regulations.

Manipulation of the injection or control system can influence the engine performance and exhaust emissions. As a result, compliance with environmental regulations may no longer be guaranteed.

Only fuels of the specified quality required to achieve emission limits must be used.

In Germany, the VAwS (=regulations governing the use of materials that may affect water quality) is applicable, which means work must only be carried out by authorised specialist companies (MTU is such a company).

Auxiliary materials

Use only fluids and lubricants that have been tested and approved by MTU.

Fluids and lubricants must be kept in suitable, properly designated containers. When using fluids, lubricants and other chemical substances, follow the safety instructions applicable to the product. Take care when handling hot, chilled or caustic materials. When using inflammable materials, avoid all sparks and do not smoke.

⇒ *Lead*

When working with lead or lead-containing pastes, avoid direct contact to the skin and do not inhale lead vapours.

Adopt suitable measures to avoid the formation of lead dust!

Switch on fume extraction system.

After coming into contact with lead or lead-containing materials, wash hands!

⇒ *Acids and alkaline solutions*

When working with acids and alkalis, wear protective goggles or face mask, gloves and protective clothing.

Immediately remove clothing wetted by acids and alkalis!

Rinse injuries with plenty of clean water!

Rinse eyes immediately with eyedrops or clean tap water.

⇒ *Painting*

When painting in other than spray booths equipped with extractors, ensure good ventilation. Make sure that adjacent work areas are not affected.

It is absolutely essential to wear masks providing protection against paint and solvent fumes.

Observe fire prevention regulations!

No smoking.

No naked flame!

⇒ *Liquid oxygen*

Liquid oxygen is highly inflammable.

Liquid oxygen should only be stored in small quantities and in regulation containers (without fixed seals)! Do not bring liquid oxygen into contact with the body (hands), as this causes frostbite and possibly the loss of tissue.

No smoking, no naked flame (risk of explosion)! Excessive oxygen in the air leads to explosive combustion.

Do not store combustible substances (e.g. oils and greases) within 5 m of the working area!

Under no circumstances should clothing be oily or greasy.

Do not allow vapours to penetrate clothing! Oxygen enrichment in fabric can cause working clothes to ignite suddenly!

After working with liquid oxygen, do not smoke until clothing is free of vapours!

Take great care to avoid impact and shock when working with liquid oxygen!

⇒ *Liquid nitrogen*

Store liquid nitrogen only in small quantities and always in regulation containers without fixed covers.

Do not bring liquid nitrogen into contact with the body (eyes, hands), as this causes frostbite and numbing.

Wear protective clothing (including gloves and closed shoes) and protective goggles!

Ensure the room is well ventilated (88% contamination of breathing with nitrogen will result in suffocation).

Avoid all knocks and jars to the containers, fixtures or workpieces.

⇒ *Compressed air*

Compressed air is air which has been compressed at excess pressure and is stored in tanks from which it can then be extracted.

The pressure at which the air is kept can be read off at pressure gauges which must be connected to the compressed air tanks and the compressed air lines.

When working with compressed air, safety precautions must be constantly observed:

- Pay special attention to the pressure level in the compressed air network and pressure vessel!
- Connecting devices and equipment must either be designed for this pressure or, if the permitted pressure for the connecting elements is lower than the pressure required, a pressure reducing valve and safety valve (set to permitted pressure) must form an intermediate connection. Hose coupling and connections must be securely attached!
- Always wear protective goggles when blowing off tools or extracting chips!
- The snout of the air nozzle is provided with a protective disc (e.g. rubber disc), which prevents air-borne particles being reflected and thereby prevents injury to eyes.
- First shut off compressed air lines before compressed air equipment is disconnected from the supply line or before equipment or tool is to be replaced!
- Unauthorised use of compressed air, e.g. forcing flammable liquids (danger class AI, AII and B) out of containers, results in a risk of explosion!
- Forcing compressed air into thin-walled containers (e. g. containers made of tin, plastic and glass) for drying purposes or to check for leaks results in a risk of explosion!
- Do not blow dirty clothing with compressed air when being worn on the body.

Even compressed air at low pressure penetrates clothing and, if the jet is directed at the back, the air can enter the anal cavity and fatally rupture the intestines!

⇒ *Used oil*

Used oil may contain health-threatening combustion residues.

Rub barrier cream into hands!

Wash hands after contact with used oil.

Warning signs

Before putting the engine into service and before working on the engine, read and follow all warning signs. Do not paint on warning signs. Replace illegible signs.

Warning notices

Publication contains especially emphasised safety instructions which begin with one of the following signal words according to the degree of danger:

Symbols of this kind draw attention to a danger which, if not heeded, may lead to serious ***injuries*** or ***death***.

In these cases, take special care.

Before taking this product into service or carrying out repairs, read and familiarise yourself with all warnings!

Pass on all safety instructions to the operating, maintenance, repair and transport personnel.

Contents

Diesel Engine

Group A
Product Summary

The product summary gives details of the engine and contains technical data such as engine power, general engine specifications, engine weight, oil and coolant capacities, settings and operational data.

ENGINE LAYOUT, ENGINE POWER

Key to Engine Model Designation 6 R 183 TD13H

6	=	Number of cylinders
R	=	In-line engine
183	=	Engine series - 100 times displacement of one cylinder in litres
T	=	Exhaust turbocharging
D	=	Air/air charge air cooling
1	=	Railway engine
3	=	Design index
H	=	Underfloor engine design

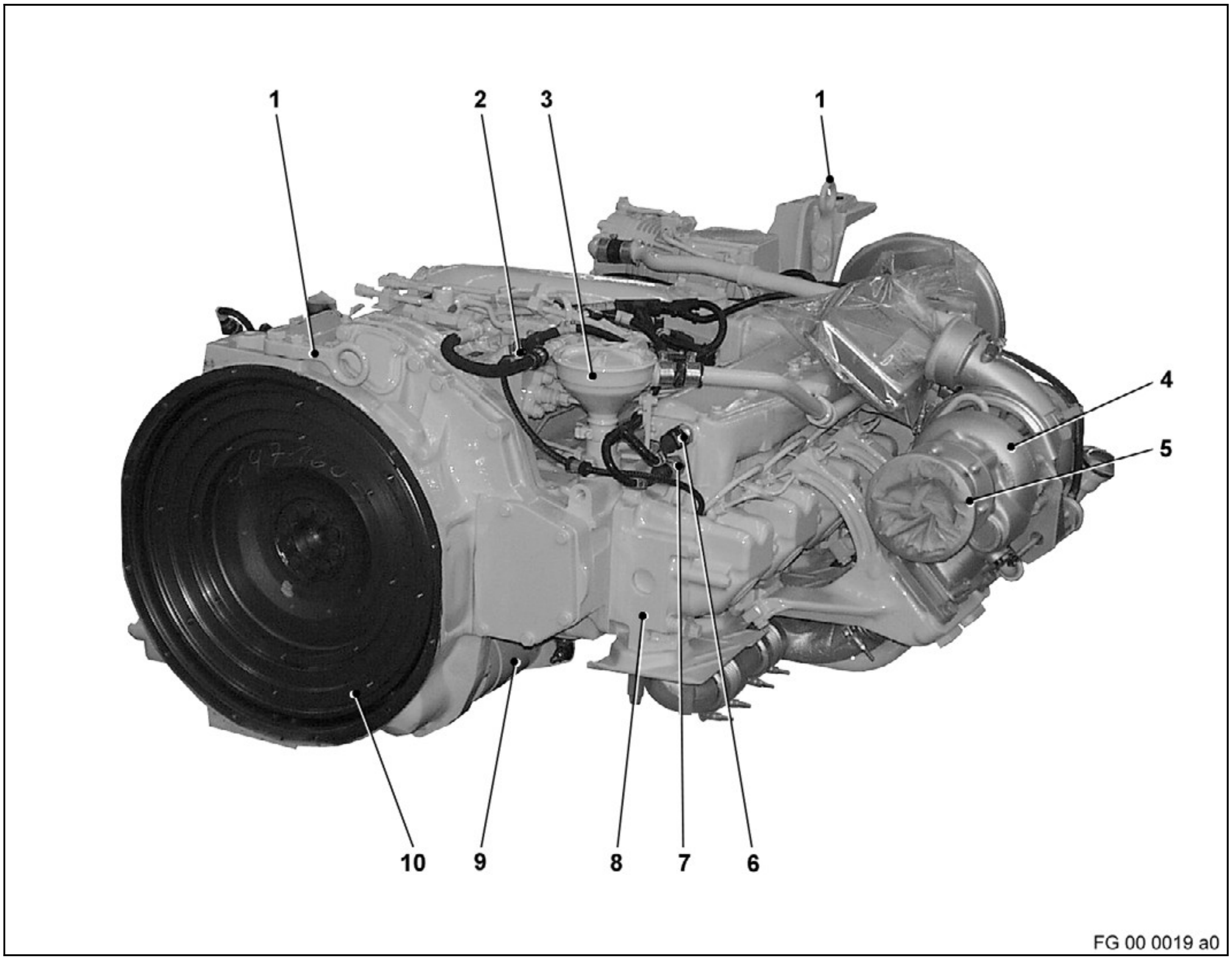
Engine Power

UIC rated power (fuel stop power)	275 kW at 1900 rpm
UIC rated power (fuel stop power)	315 kW at 1900 rpm

Reference conditions

– intake air temperature	25 °C
– barometric pressure	1000 mbar
– site altitude	100 m

ENGINE LAYOUT, DRIVING END, RIGHT SIDE



1	Lifting eye	6	Pressure transmitter (charge air pressure)
2	Temperature transmitter (fuel temperature)	7	Temperature transmitter (charge air temperature)
3	Oil separator	8	Cylinder head
4	Exhaust turbocharger	9	Electric starter
5	Exhaust pipework connection	10	Flywheel